

Machine Learning in Computational Biology Spring 2026

Assignment #2

Heart Disease Classification using Repeated Nested Cross-Validation

MSc: Data Science & Information Technologies

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GitHub repository: https://github.com/Mushroompilamis/MLCB26_Assignment2

Abstract

In this analysis, the development of a complete machine learning pipeline for binary heart disease classification using the Cleveland Heart Disease dataset was attempted, which contained 242 patient records with 13 clinical features and one target variable indicating the absence or presence of heart disease.

An exploratory data analysis was performed, and then preprocessing was incorporated inside the cross-validation pipeline to avoid data leakage. Seven classifiers were compared using repeated nested cross-validation with 10 repetitions, 5 outer folds, and 3 inner folds, with Linear Discriminant Analysis (LDA) achieving the strongest overall performance, with the highest median MCC and strong performance across ROC-AUC, balanced accuracy, F1-score, specificity, and PR-AUC. In later steps with feature selection, the identification of clinically meaningful variables such as ca, cp, thal, thalach, oldpeak, and exang occurred, although the reduced-feature model did not outperform the full-feature model.

Lastly, the final LDA model was trained on all available data and interpreted using SHAP values, which showed that vessel involvement, thalassemia category, chest pain type, maximum heart rate, and exercise-induced changes were the main drivers of prediction, with a follow-up error analysis showing that false negatives were mainly associated with less clearly pathological profiles, such as higher thalach and lower oldpeak.

Overall, the final model provided an interpretable and clinically meaningful classification pipeline.

Introduction

The Cleveland Heart Disease dataset used in this analysis (`students_dataset.csv`) was collected from the Cleveland Clinic Foundation and is one of the most widely studied benchmark datasets in medical machine learning. The dataset contains 242 patient records and 14 attributes in total, consisting of 13 clinical features and one target variable. Each record describes a patient through clinical measurements that are commonly used for the assessment of coronary artery disease.

The prediction task is formulated as a binary classification problem based on the target variable `num`. Specifically, `num = 0` corresponds to the absence of heart disease, while `num = 1` corresponds to the presence of heart disease. The 13 input features include demographic, clinical, electrocardiographic, and exercise-related variables, such as age, sex, chest pain type (`cp`), resting blood pressure (`trestbps`), serum cholesterol (`chol`), fasting blood sugar (`fbs`), resting electrocardiographic results (`restecg`), maximum heart rate achieved (`thalach`), exercise-induced angina (`exang`), ST depression induced by exercise (`oldpeak`), the slope of the peak exercise ST segment (`slope`), number of major vessels colored by fluoroscopy (`ca`), and thalassemia type (`thal`). These variables include continuous, binary, categorical, and ordinal feature types, which makes appropriate preprocessing an important part of the modelling pipeline.

Since the dataset may contain missing values and potential class imbalance, these issues must be handled carefully in order to build reliable predictive models [3]. In particular, preprocessing steps such as imputation, encoding, scaling, and feature selection must be performed inside the cross-validation pipeline, so that information from validation or test folds is not leaked into the training process [3], [8]. This is especially important in medical machine learning, where overestimated performance can lead to misleading conclusions about the clinical usefulness of a model.

Therefore, the aim of this analysis was to develop a complete machine learning pipeline for heart disease classification using repeated nested cross-validation [1]. The analysis started with an exploratory investigation of the UCI Heart Disease dataset, focusing on dataset structure, missing values, feature distributions, class balance, and potential relationships between features and the target variable. Then, an abstraction of the nested cross-validation logic was designed in order to clarify the flow of data through preprocessing, model training, hyperparameter tuning, and outer-fold evaluation.

Following this, a repeated nested cross-validation pipeline was implemented using object-oriented programming principles [13]. Several classification algorithms were evaluated and compared in order to identify the model that achieved the best balance of clinically relevant performance metrics [1]. After selecting the winning algorithm, feature selection was performed to investigate whether a reduced subset of informative variables could maintain or improve predictive performance. Finally, the selected model was trained on all available data and saved as a complete deployable pipeline. Model interpretation was performed using SHAP values, allowing both global and local explanations of the model's predictions. In addition, error analysis was used to examine false positive and false negative predictions and to identify patient characteristics associated with misclassification.

Materials & Methods

Dataset & repository structure

In this analysis the dataset of use students_dataset.csv, was the Cleveland Heart Disease, which contains 242 patient records and 14 columns in total, corresponding to 13 clinical features and one target variable, with the prediction task formulated as a binary classification problem, where the target variable num indicates the absence (num = 0) or presence (num = 1) of heart disease. Regarding the variety of the input features, include continuous clinical measurements, binary variables, categorical variables, and ordinal variables, which make appropriate preprocessing necessary before model training.

The project repository of this analysis was organized into separate folders in order to keep the analysis reproducible and structured, with the data/ directory containing the raw dataset [10]. In the figures/ directory, the figures, which are produced during exploratory analysis and model evaluation, are stored. Also, a models/ directory is created, which is reserved for the final trained model instance, which will be saved as a complete deployable pipeline. More directories such as notebooks and src are created, with the first one containing the notebooks corresponding to the assignment tasks, including exploratory data analysis, repeated nested cross-validation, feature selection, final model training, SHAP interpretation, and error analysis, and the second, final one containing the Python source code, including the script that implements the repeated nested cross-validation class used for model evaluation.

Exploratory Data Analysis

The exploratory data analysis step was performed in the notebook 01_EDA.ipynb, aiming to understand the structure of the dataset, to identify possible data quality issues, and to obtain an initial view of the relationships between the clinical features and the target variable before model development.

First, the dataset was loaded using pandas, and its shape, column data types, and descriptive statistics were examined in order to confirm the number of patient records and variables, to inspect the format of the dataset, and to check whether the features were loaded with appropriate numerical types. Descriptive statistics were also used to examine the range, central tendency, and spread of the numerical features [4].

The dataset was then checked for duplicate rows and missing values, with missingness to be quantified both as counts and percentages for each feature. Notably, missing values were not imputed globally during EDA because this could introduce data leakage if statistics from the full dataset were later used during cross-validation. Instead, imputation was planned to be performed later inside the cross-validation pipeline. Furthermore, since it was observed that the missing values appeared only in ca and thal, median/mode imputation was considered appropriate according to feature type.

Additionally, potential outliers were examined only for continuous features using the interquartile range method, assuming a meaningful numerical scale [2], with the continuous variables being considered age, trestbps, chol, thalach, and oldpeak. Regarding the binary, the categorical, and the ordinal variables, they were inspected through their category distributions. The resulting potential outliers, though, were not removed at this stage, because extreme clinical measurements may correspond to real patient profiles rather than data entry errors, such as high cholesterol.

A next step in data inspection, class distribution was also examined to assess whether class imbalance was present, since class imbalance affects both model training and evaluation [11],

with this step justifying the later use of stratified cross-validation and complementary metrics such as MCC, balanced accuracy, recall, specificity, ROC-AUC, PR-AUC, precision, and F1-score [11].

For feature assessment, boxplots were created for the continuous variables grouped by the target class, plots that were used to compare the distributions of continuous clinical measurements between patients with and without heart disease. For the binary, the categorical, and the ordinal variables, stacked bar plots were created, since their numerical values represent discrete categories rather than continuous measurements.

A Spearman correlation heatmap was also generated to examine monotonic associations between the features and the target variable, with Spearman correlation being more preferred over Pearson correlation because the dataset includes binary, ordinal, and categorically encoded clinical variables [4]. The previously found missing values were temporarily imputed only for visualization purposes in this step, and the resulting imputed dataset was not used for model training, with the final correlation heatmap being interpreted only as an exploratory tool and not as a criterion for feature removal.

Finally, PCA was used as a dimensionality reduction technique to visualize whether the two classes showed visible separation in a two-dimensional projection of the feature space [4]. For running PCA, temporary preprocessing was applied only for visualization and not for the machine learning pipeline, with the continuous features being median-imputed and standardized, the categorical features mode-imputed and one-hot encoded, and the binary/ordinal features imputed according to their type.

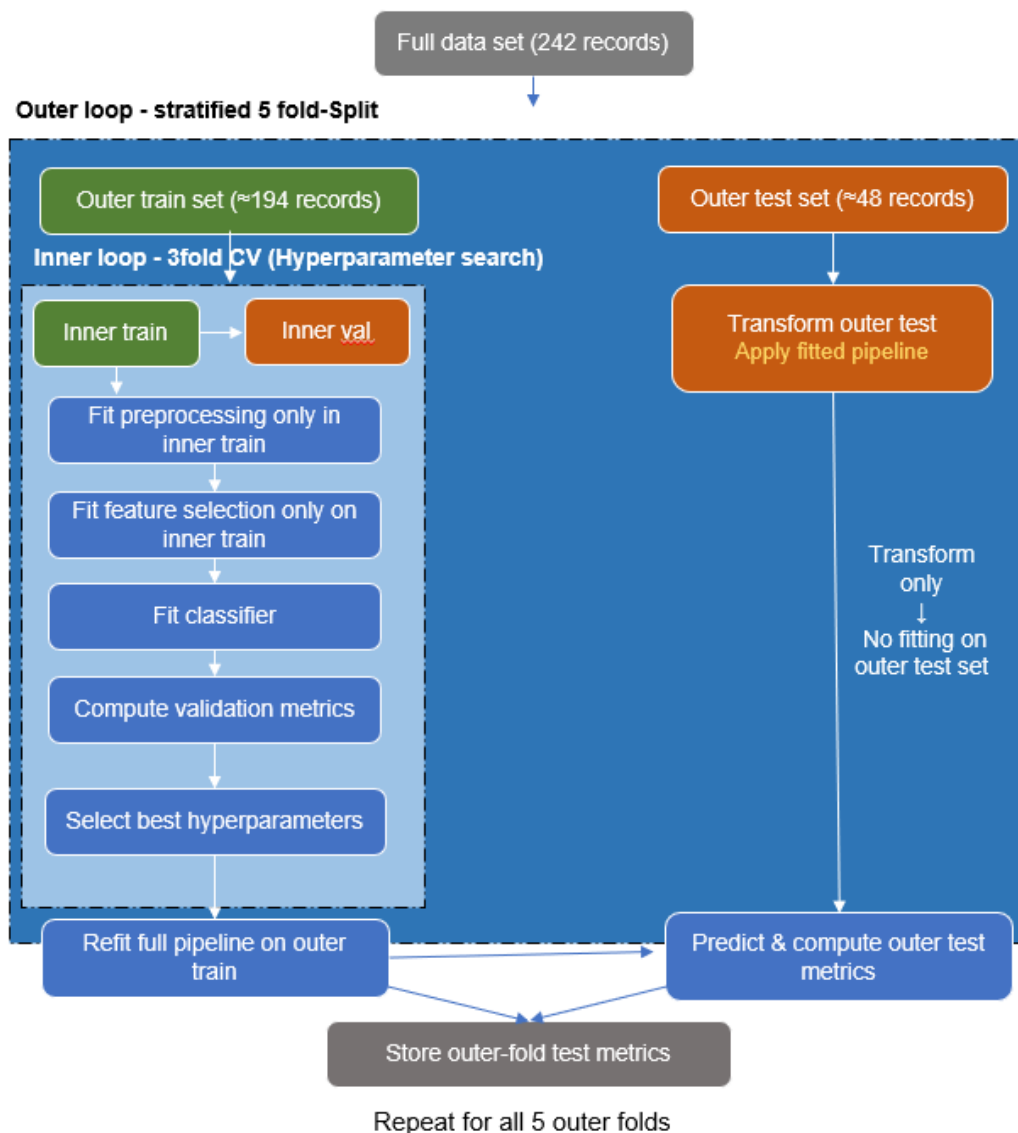
Intuition & Preparation for the nCV Implementation

The main goal of the analysis consists of the implementation of a nested cross-validation pipeline following object-oriented programming principles. Although, before implementation, the repeated nested an abstraction of one nested cross-validation iteration was designed in order to clarify the flow of the data through the training, tuning, and evaluation steps by defining where preprocessing, feature selection, hyperparameter tuning, and final testing should occur while avoiding data leakage.

As shown in Figure 1, the full dataset was first split using an outer stratified 5-fold cross-validation scheme, with stratification being used in order to preserve the class distribution of the target variable across folds. In one outer-fold iteration, the data are separated into an outer training set, containing approximately 194 records, and an outer test set, containing approximately 48 records, with the second one remaining locked during all preprocessing, feature selection, and hyperparameter tuning steps.

The outer training set is then passed to an inner stratified 3-fold cross-validation loop, which is used for hyperparameter selection. Within each inner split, preprocessing is fitted only on the inner training fold and then applied to the corresponding inner validation fold. The same logic applies to feature selection, which is fitted only on the inner training fold. The classifier is then trained on the processed inner training data and evaluated on the inner validation fold, which is used to select the best hyperparameter configuration.

After the best hyperparameters are selected, the complete pipeline is refitted on the full outer training set, including preprocessing, optional feature selection, and classifier training.



Repeated Nested Cross-Validation implementation

number of repetitions, the number of outer and inner folds, the number of optimization trials, the random seed, and whether hyperparameter tuning or feature selection should be enabled.

Before model training, the features were separated according to their data type, because each type requires different preprocessing. Regarding each category of variables, the continuous variables were median-imputed and standardized; the binary variables, including sex, fbs, and exang, were imputed using the most frequent value and kept in their original 0/1 representation; the categorical variables, including cp, restecg, and thal, were imputed using the most frequent category and one-hot encoded so that their numerical codes would not be treated as continuous values; and lastly, the ordinal variables, including slope and ca, were median-imputed and kept as ordered numerical variables.

In addition, the preprocessing steps were implemented using a ColumnTransformer, which allowed each group of features to be processed with the appropriate pipeline [8]. The preprocessor was then placed inside a scikit-learn pipeline together with the classifier [8], and as a result, imputation, scaling, and encoding were fitted only on the training part of each fold and then applied to the corresponding validation or test folds, aiming to avoid data leakage.

On the one side, the outer loop of the `rnCV` class used stratified K-fold cross-validation to preserve the class distribution of the target variable across folds; within each outer fold, the data were split into an outer training set and an outer test set, which was used only for the final evaluation of that fold.

On the other side, the inner loop was applied only on the outer training set and was used for hyperparameter optimization. Optuna, specifically, was used to search the hyperparameter space of each classifier [6],[7],[14] while for every Optuna trial, a complete preprocessing-classifier pipeline was evaluated using inner stratified K-fold cross-validation. Additionally, the mean inner validation score was used to select the best hyperparameter configuration, and ROC-AUC was used in this implementation as the optimization metric in the inner loop.

After the best hyperparameters were selected, the complete pipeline was refitted on the full outer training set. The fitted pipeline was then applied to the outer test set to obtain predictions and positive-class probability scores, predictions that were used to compute several evaluation metrics, such as Matthews Correlation Coefficient, ROC-AUC, balanced accuracy, F1-score, recall, specificity, precision, and precision-recall AUC, metrics that were stored for every outer fold and every repetition.

Finally, in the class was created a summary function, which aggregates the outer-fold results for each algorithm, with the median of each metric being used as the main performance estimate since the repeated nested cross-validation procedure produces 50 outer test evaluations per algorithm. In addition, 95% bootstrap confidence intervals were calculated for the median metrics in order to quantify uncertainty in the estimated generalization performance [1], [3].

Model selection and `rnCV`

The next step was the model selection analysis (notebooks/03_model_selection.ipynb), with the dataset loaded, the target variable num converted into binary format, and the feature matrix X defined by removing the target column. Under the same framework, 7 classification algorithms were evaluated, which were Logistic Regression with Elastic Net regularization, Gaussian Naive Bayes, Linear Discriminant Analysis, Random Forest, LightGBM, XGBoost, and CatBoost; models which were selected in order to compare linear, probabilistic, discriminant-analysis, bagging-based, and boosting-based approaches.

First, a baseline comparison was performed by disabling the inner hyperparameter tuning loop, where all algorithms were evaluated using their default or predefined hyperparameters over 10 repetitions of stratified 5-fold cross-validation, producing 50 outer-fold evaluations per algorithm, providing an initial fair comparison before hyperparameter optimization.

After the baseline analysis, algorithm-specific Optuna search spaces were defined, and the full rnCV procedure was then performed using the custom RepeatedNestedCV class [7], [14]. The outer loop consisted of stratified 5-fold cross-validation repeated 10 times and was used to estimate generalization performance [5]. The inner loop consisted of stratified 3-fold cross-validation applied only on the outer training set and was used for hyperparameter tuning [5]. For each algorithm, Optuna evaluated up to 50 trials per outer fold, using ROC-AUC as the optimization metric [7], [14].

For each outer fold, the best hyperparameters selected from the inner loop were used to refit the complete preprocessing-classifier pipeline on the full outer training set [5]. The fitted pipeline was then evaluated once on the untouched outer test set [5].

Lastly, the 7 algorithms were compared using the median performance across the 50 outer-test evaluations, with the Median MCC to be used as the primary ranking metric because it summarizes binary classification performance using all entries of the confusion matrix [9]. Furthermore, ROC-AUC, balanced accuracy, F1-score, recall, specificity, precision, and PR-AUC were also examined to assess performance from multiple clinically relevant perspectives [9], [11]. Also, for each metric, 95% bootstrap confidence intervals were calculated for the median estimates, and the algorithm with the strongest median MCC and a clinically reasonable balance across the remaining metrics was selected as the winner for downstream analysis [9].

Feature selection

After the winner algorithm that had been selected from the full rnCV comparison, the feature selection was evaluated, aiming to examine whether a reduced and more stable subset of clinical features could preserve the performance of the full-feature model while improving interpretability.

Feature selection was performed using a mutual-information-based filter method through SelectKBest, because it is model-independent and can capture non-linear associations between each feature and the target variable. The step of selection, specifically, was enabled inside the custom RepeatedNestedCV class and was applied only within the training data of each outer fold to avoid data leakage.

Additionally, the rnCV procedure was rerun only for the winner model, with the same structure as before, while for each outer fold, preprocessing was fitted on the outer training set, mutual-information scores were computed on the preprocessed training data, and the selected features were then used to train and evaluate the model on the corresponding outer test set.

However, because the categorical variables were one-hot encoded during preprocessing, feature selection was first produced at the transformed-feature level. Therefore, one-hot

encoded categories were aggregated back to their original clinical variables to improve interpretation. Also, selection frequencies were calculated across all outer-fold evaluations, and features selected in at least 80% of the folds were considered stable.

Finally, the winner model was evaluated again using only the stable original features, since the purpose was to test the performance of the already selected stable feature subset against the full-feature model.

This choice allowed the comparison between the full model and the reduced model under the same rncv framework.

Final model training & SHAP interpretation

By obtaining the winner algorithm, one more final step, a tuning step, was performed to determine the hyperparameters used for the deployable model. A stratified 5-fold cross-validation procedure was also applied to the full dataset using Optuna, with ROC-AUC as the optimization metric, with the same hyperparameter space used during model selection reused for the winner model to keep the final tuning procedure consistent [7], [14].

Using the selected hyperparameters, the final model was trained on all available data and saved as a complete scikit-learn pipeline, including both preprocessing and the classifier, a choice useful for direct acceptance of raw patient data and application of the required imputation, scaling, encoding, and classification steps in the correct order.

Using that final training model, a SHAP analysis was then used for interpretation. Since the winner model was LDA, SHAP values were computed using LinearExplainer, which is appropriate for linear models, but before computing SHAP values, the dataset was transformed using the fitted preprocessing step of the final pipeline because the classifier operates on the preprocessed feature space rather than on the raw input variables [15].

More specifically, the SHAP analysis was performed at the transformed-feature level, in which the categorical variables that were one-hot encoded during preprocessing appeared as category-specific dummy variables [15]. For visual inspection of the results also, both a SHAP beeswarm plot and a mean absolute SHAP importance plot were used to observe the direction of each feature's effect on the model output, while the mean absolute SHAP plot was used to rank features according to their average contribution magnitude.

Error analysis

As a last step of the analysis, a validation-based error analysis was performed to better understand the types of mistakes made by the selected model.

Firstly, the dataset was split into a training set and a validation set (80/20 stratified split) so that the class distribution was preserved in both subsets [3], with the LDA model to be trained on the training set using the selected final hyperparameters.

Then, validation predictions were classified into four groups (true positives: TP, true negatives: TN, false positives: FP, and false negatives: FN), with a confusion matrix and a classification report to be used to summarize the model's validation performance.

Specifically, the group of the FN was considered especially important in this clinical context because they correspond to patients with heart disease who were predicted as not having the disease.

Therefore, to examine whether misclassifications were associated with specific patient profiles, selected clinical features were compared across prediction outcome groups, with Median feature values to be computed for each group, and normalized feature-profile heatmaps to be used to highlight differences between correctly classified and misclassified cases.

SHAP values were also computed on the validation set using the trained error-analysis pipeline, with the validation data first being transformed using the fitted preprocessor, since the model operates on the preprocessed feature space [15]. In addition, Mean absolute SHAP values were then compared between FP and FN cases, as well as between correctly classified and misclassified cases, to identify which features contributed most strongly to the observed prediction errors.

Results & Discussion

Exploratory data analysis results

The exploratory data analysis showed that the dataset was relatively small, with 242 patient records and a slight class imbalance (1.18) between the no-disease and disease groups (Figure 2A). Missing values were also inspected and seemed to be limited to *ca* and *thal*, with only one missing value in each feature. These features were not imputed during EDA but were handled later inside the cross-validation pipeline to avoid data leakage. Regarding the outlier analysis, it showed several potential extreme values, especially in continuous clinical variables such as *trestbps* and *chol* (Figure 2C), but these values were retained since they represent real clinical profiles.

From the feature visualizations, it was indicated that several variables may be informative for classification. Continuous features such as *thalach*, *oldpeak*, and *age* showed visible differences between the two target classes (Figure 2C), while categorical and ordinal features such as *cp*, *ca*, *thal*, *exang*, and *slope* also showed class-related patterns (Figure 2D). Also, with a Spearman correlation heatmap, we supported these findings, with *thal*, *ca*, *cp*, and *exang* showing positive associations with disease status and *thalach* showing a negative association (Figure 2E). Finally, with PCA it was shown only a partial separation between classes (Figure 2B), indicating that the classification task likely requires the combined information of multiple features rather than a simple two-dimensional separation.

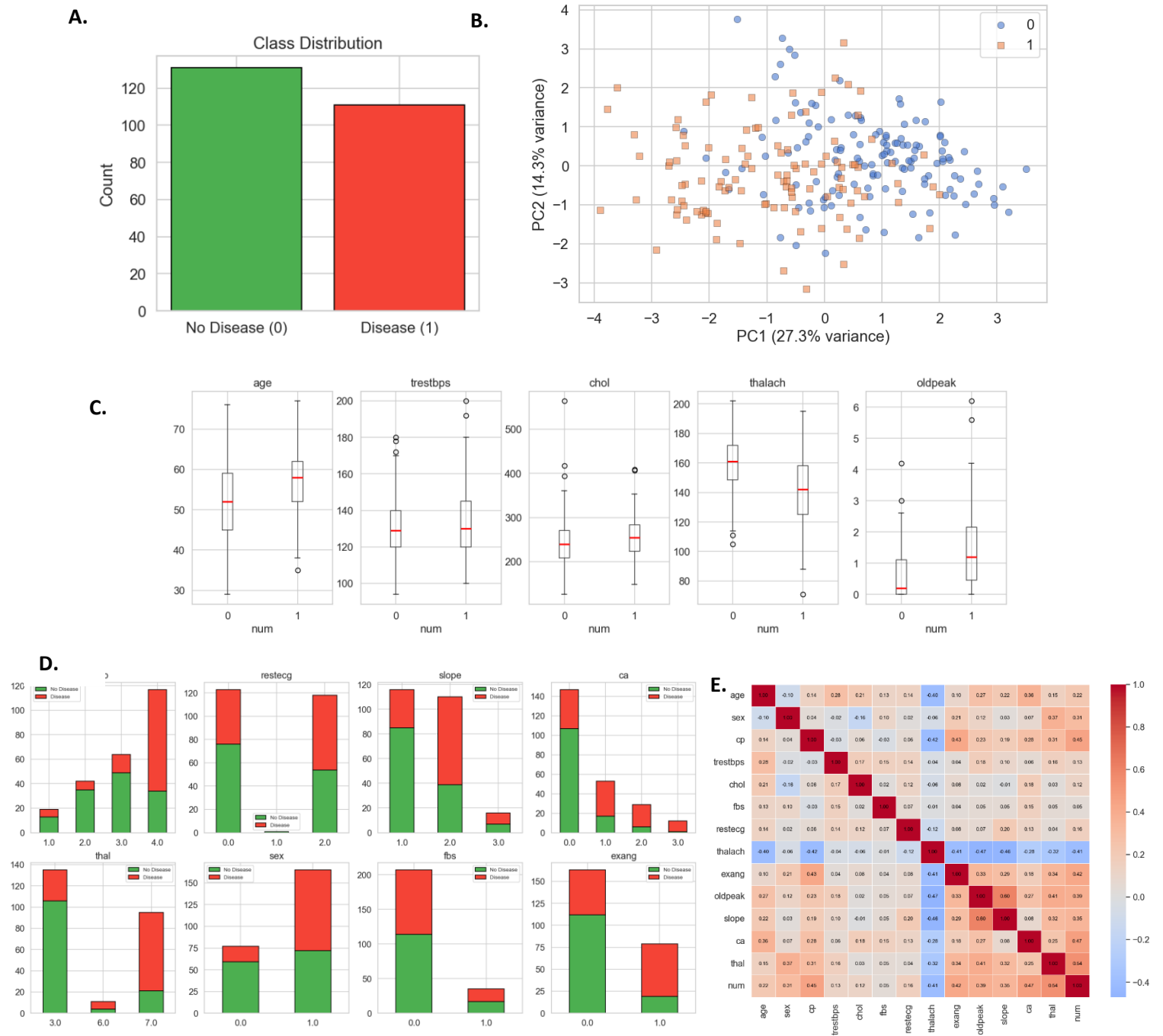


Figure 2: A) Class distribution of the target variable. Number of patients without heart disease (0) and with heart disease (1). B) PCA projection of the feature space, into two-dimensional projection of the preprocessed features using the first two principal components. C) Box plots of continuous features by disease status. Distribution of continuous clinical features grouped by target variable. D) Stacked bar plots of categorical, binary, and ordinal features by disease status. E) Spearman correlation heatmap, with pairwise monotonic associations between features and the target variable [1].

rnCV

Regarding the baseline comparison, it was shown that LDA and Logistic Regression were the strongest untuned models (Figure 3A), with LDA achieving the highest median MCC (0.6643). After also repeated nested cross-validation with Optuna tuning, LDA remained the top-performing model, achieving a median MCC of 0.6643 with a 95% bootstrap confidence interval of [0.6296, 0.6751] (Figure 3B). It also showed strong median AUC (0.8974), balanced accuracy (0.8217), F1-score (0.7907), and PR-AUC (0.8906) (Figure 3C). On the contrary, Gaussian Naive Bayes before and after tuning showed the lowest median MCC and the largest variability, suggesting less stable performance across folds (Figures 3A,3B,3C). Therefore, LDA was selected as the winner algorithm for the subsequent feature selection, final model training, and interpretation steps.

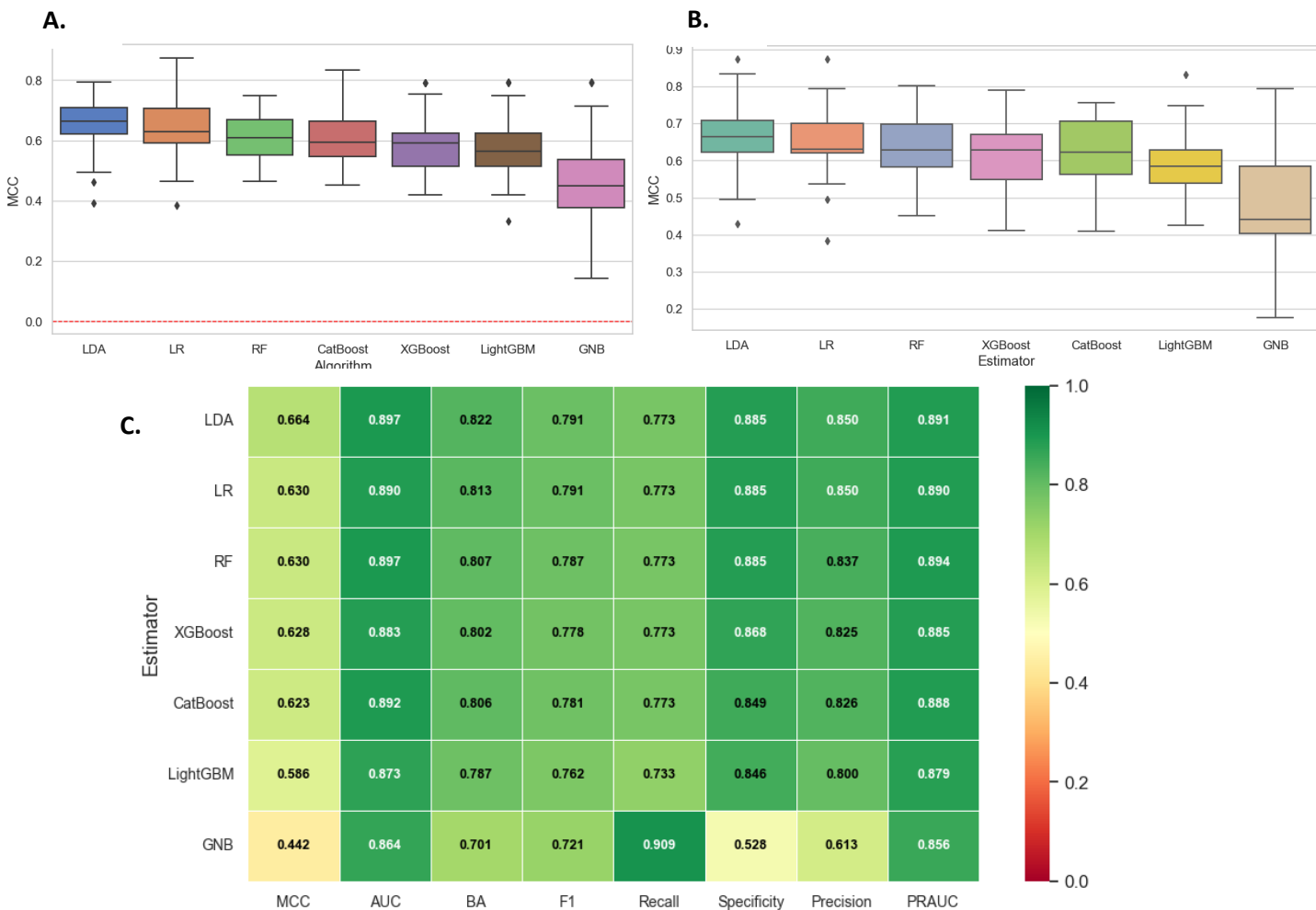


Figure 3: A) Baseline MCC distributions across the algorithms with MCC values specifically across 50 outer-fold evaluations for each classifier without hyperparameter tuning. B) Distribution of MCC values across the 50 outer-test evaluations after hyperparameter tuning. C) Multi-metric heatmap of tuned model performance, with median performance metrics for each algorithm after repeated nested cross-validation, ordered by median MCC [1].

Feature selection & Hyperparameter tuning

During feature selection, six stable features were identified and selected in at least 80% of the outer-fold evaluations: ca, cp, thal, thalach, oldpeak, and exang (Figure 4A). When the winner model was re-evaluated using feature selection and using only the stable selected features, the performance remained essentially unchanged compared with the full-feature LDA model. The full LDA model achieved a median MCC of 0.6643, while the feature-selection and stable-feature-only models [8] also achieved a median MCC of 0.6643 (Figure 2B).

This suggests that the stable feature subset preserved most of the predictive information needed by the LDA classifier. Therefore, the reduced feature set may be useful for interpretability, since it focuses on clinically meaningful variables while maintaining similar classification performance. However, the full-feature model was still retained for the final deployed pipeline, because it used all available clinical information and showed slightly stronger AUC compared with the stable-feature-only model.

Additionally, after LDA was selected as the winner model, final hyperparameter tuning was performed using stratified 5-fold cross-validation on the full dataset [12]. The best configuration was solver = lsqr and shrinkage = 0.3313, with a mean CV ROC-AUC of 0.9058. The final model was then trained on all available data and saved as a complete preprocessing-classifier pipeline. With SHAP interpretation also, it was shown that the model was mainly driven by ca, thal, cp, thalach, exang, slope, and oldpeak (Figure C). Although, since categorical variables were one-hot encoded, such as thal and cp, the SHAP values for the features were interpreted at the category-specific dummy-variable level. However, a bar plot of the mean SHAP was created, indicating that ca has the strongest average impact on the final LDA model output (Figure D).

Overall, the SHAP results were consistent with the EDA and feature selection findings.

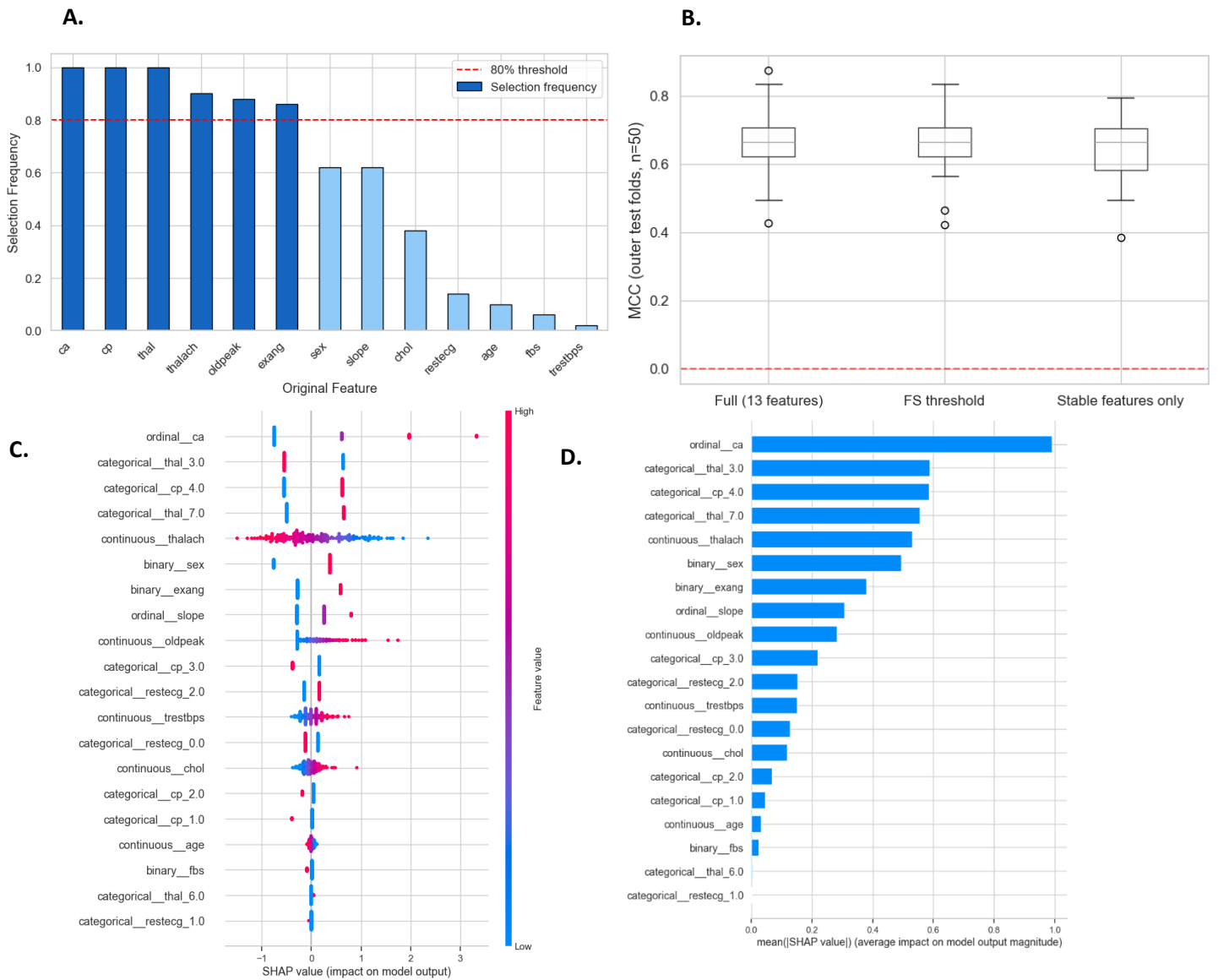


Figure 4: A) Bar plot depicting the selection frequency of each original clinical feature across the repeated nested cross-validation procedure, with the red dashed line indicating the 80% stability threshold. B) Distribution of MCC scores across 50 outer-test evaluations for the full LDA model, the feature-selection pipeline, and the stable-feature-only model. C) SHAP beeswarm plot for the final LDA model, showing the direction and magnitude of each preprocessed feature's contribution to the model output. D) Mean absolute SHAP importance for the final LDA model [1].

Error analysis

Regarding the error analysis, it showed that the LDA model achieved strong validation performance, with 0.90 accuracy on the validation set, with the model correctly classifying 26 no-disease patients as true negatives and 18 diseased patients as true positives, while producing 1 false positive and 4 false negatives (Figure 5A). The disease-class recall was 0.82, meaning that some heart disease cases were still missed, making those false negatives clinically important, since they correspond to patients with heart disease who were predicted as healthy.

As for the clinical profile analysis, it suggested that false negative cases had less clearly pathological patterns compared with true positives. Specifically, false negatives showed higher median thalach and lower median oldpeak, together with lower chol, making them appear more similar to no-disease cases (Figure 5C). In contrast, true positives showed higher median chol, higher oldpeak, higher ca, and lower thalach (Figure 5C), which are patterns more strongly associated with heart disease and may explain why these cases were easier for the model to identify correctly.

Additionally, age alone did not appear to clearly explain the model errors, since the age distributions of correct and incorrect predictions were broadly overlapping (Figure 5B), suggesting that the misclassification was more likely related to combinations of clinical variables rather than to a single demographic feature.

Also, the SHAP-based error analysis further supported this interpretation. For both false positives and false negatives, the largest mean SHAP contributions mainly came from `ca`, `thal`, `cp`, `exang`, and `slope`, which were also among the main drivers of the overall model predictions (Figure 5E), indicating that the errors were not random but were mostly associated with the same clinically meaningful variables used by the model to make correct predictions. However, the comparison between correctly classified and misclassified patients showed smaller or different SHAP contributions for some features in the misclassified group, suggesting that these patients may have had more ambiguous or mixed clinical profiles (Figure 5D).

Overall, false negatives remain the most clinically important limitation of the model because they represent disease cases whose available feature patterns were not strong enough for the model to classify them correctly, and that might affect feature diagnoses.

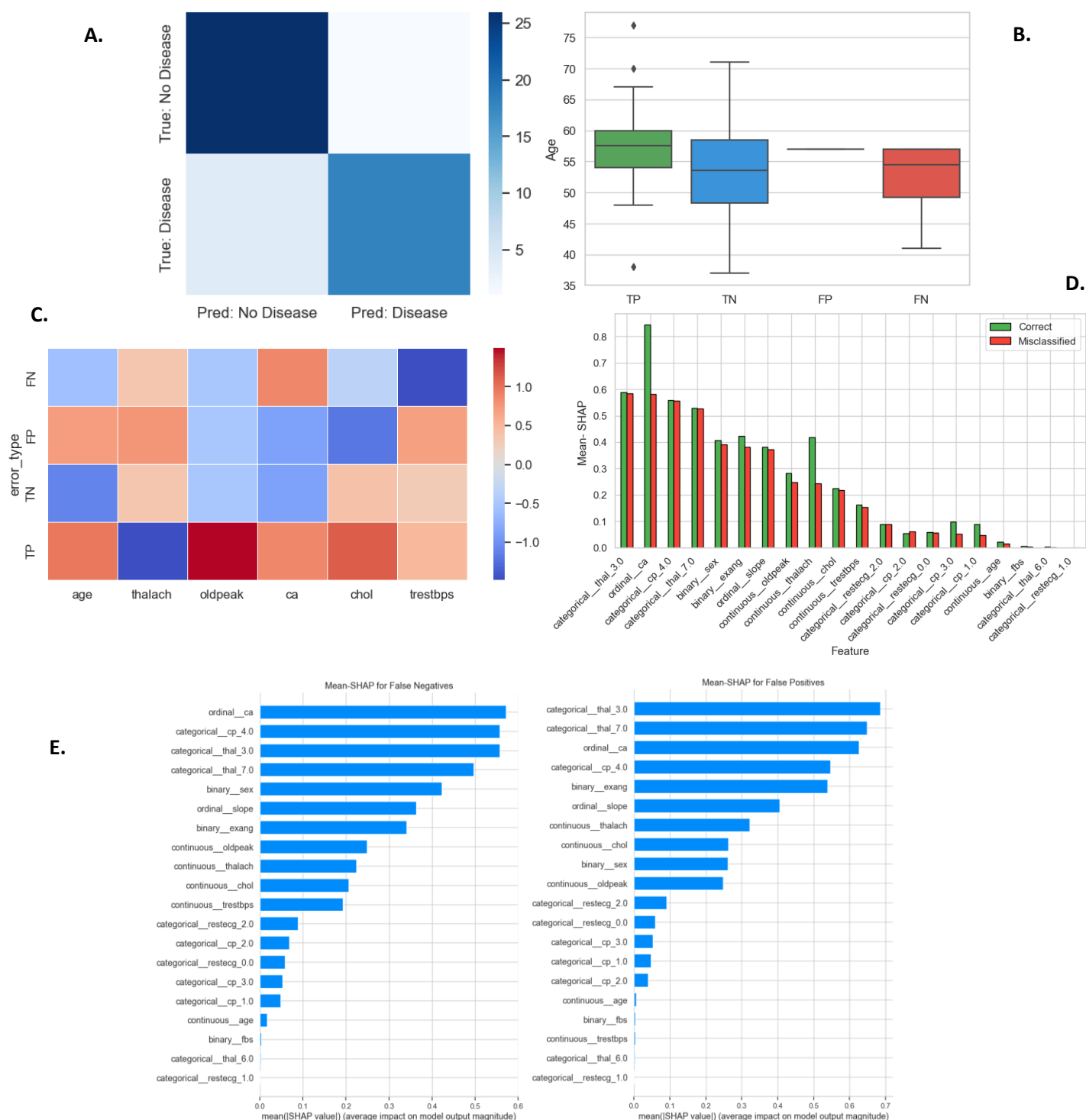


Figure 5: A) Confusion matrix on the validation set. Validation predictions of the final LDA model separated into true negatives, true positives, false positives, and false negatives. B) Box plot age distributions of true positives, true negatives, false positives, and false negatives in the validation set. C) Normalized clinical feature profiles by prediction outcome. Median feature profiles for TP, TN, FP, and FN cases, normalized across outcome groups. D) SHAP comparison between correctly classified and misclassified patients. E) Mean absolute SHAP values for false negative and false positive cases [1].

Conclusions

The first problem that we had to encounter was that the dataset was relatively small, containing only 242 patient records; however, it included clinically diverse information from continuous, binary, categorical, and ordinal variables, which made preprocessing an important part of the analysis, especially because missing values, different feature scales, and categorical encodings had to be handled carefully inside the cross-validation pipeline.

During the exploratory data analysis step, it was shown that several features had visible relationships with the target variable, such as thalach, oldpeak, cp, ca, thal, and exang, which appeared to contain useful predictive information. Specifically, patients with heart disease tended to have lower maximum heart rate achieved during exercise, higher ST depression, more exercise-induced angina, and different patterns in chest pain type, thalassemia category, and number of major vessels, findings which were also supported by the correlation analysis and later by the SHAP interpretation.

During the model comparison, which was performed using repeated nested cross-validation, it provided a more reliable estimate of generalization performance than a single train-test split, with the baseline comparison already suggesting that linear models were competitive for this dataset. Additionally, after hyperparameter tuning, Linear Discriminant Analysis achieved the best overall performance, with the highest median MCC and strong values across AUC, balanced accuracy, F1-score, specificity, and PR-AUC, meaning that despite the presence of some non-linear clinical relationships, the dataset may contain a feature structure that can be effectively separated using a regularized linear discriminant approach [12].

Regarding feature selection, it showed that several variables were selected consistently across folds, including clinically meaningful features such as ca, cp, thal, thalach, oldpeak, and exang. However, when the model was re-evaluated using only the reduced stable feature subset, performance decreased compared with the full-feature model, indicating that although some features are clearly more important than others, useful predictive information may still be distributed across the full set of 13 variables. Therefore, the full-feature LDA model was retained as the final model.

As for the SHAP analysis, it provided additional support for the model interpretation. The most influential features included ca, thal, cp, thalach, exang, slope, and oldpeak, which agree with both the EDA and feature selection results. Higher ca, oldpeak, and exang values generally pushed predictions toward the heart disease class, while higher thalach pushed predictions toward the no-disease class, although since categorical variables were one-hot encoded, some SHAP effects were interpreted at the category-specific level rather than at the original raw-variable level.

Also, the error analysis showed that the final model performed well on the validation set but still produced clinically important false negative patients, which appeared to have less clearly pathological profiles, such as higher thalach and lower oldpeak, making them more similar to no-disease cases, suggesting in this way that the model may struggle with patients whose clinical patterns are mixed or less typical. From a clinical perspective, this is important since missing a true disease case may have more serious consequences than incorrectly flagging a healthy patient.

Overall, the results show that the final LDA model provides a strong and interpretable baseline for heart disease classification. The consistency between EDA, feature selection, SHAP interpretation, and error analysis increases confidence that the model is using clinically meaningful signals rather than arbitrary patterns. However, the small dataset size and the limited number of validation errors limit the strength of the conclusions, with future work being mandatory to evaluate the model on larger and more diverse external cohorts and aim to reduce false negatives in clinically sensitive settings.

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AI Usage Disclosure

Large Language Model tools were used during the preparation of this assignment, such as ChatGPT or QuillBot. The first category of use was as an assistant for code debugging, formatting corrections, understanding the machine learning methodology, and clarifying the logic behind the repeated nested cross-validation implementation. It was also used to support plotting choices, figure formatting, coloring, and improved visual representation of the results. In addition, the second category was used to improve the clarity, grammar, and readability of parts of the written report.

Although, each final decision regarding the implementation, interpretation of results, and scientific content was reviewed and validated by the author.